

Engineering Reboot 2026

Day 25 Supplemental Guide

Purpose

This guide is a concise engineering briefing to accompany the official Day 25 workbook. It explains what each exercise expects, suggests an implementation approach, and provides expected results without replacing the workbook.

Exercise Format

Each exercise contains: **Objective** — What you are learning. **Requirements** — Exactly what must be implemented. **Suggested Implementation Approach** — Recommended sequence of work. **Expected Output** — What you should see when complete. **Engineering Notes** — Common pitfalls and best practices.

Mini Project Format

The mini project section should contain: Functional requirements. Recommended program structure. Reusable functions from Days 23–24 to leverage. Suggested testing checklist. Success criteria before committing the project.

Example – Exercise 3

Objective

Modify persisted employee data and verify that the change survives program termination.

Requirements Load *employees.json*. Locate employee ID 1005. Increase the salary by 5%. Save the updated data. Run the program again and confirm the new salary is still present. **Suggested Approach** Load the JSON file into a list of dictionaries. Find employee 1005. Multiply the salary by 1.05. Save using `json.dump(..., indent=4)`. Reload the file to verify persistence. **Expected Result** The salary is updated in *employees.json*. Restarting the application displays the updated salary.

Engineering Note

This exercise introduces persistent application state through JSON storage.